

1. Assume that the probability of rain today is E, and tomorrow is F.

$$P = P(E) + P(F) + P(E \cap F) = 9/10 * 9/20 + 1/10 * 11/20 + 9/10 * 11/20 = 191/200 = 95.5\%$$

2. (a): Assume that the probability for which the first card is red is E, and black is F:

$$P = P(E) + P(F) = 1/13 * 1/2 = 1/26$$

- (b): Assume that the probability of the first card is E, and the second is F:

$$P = P(E \cap F) = P(E) * P(F) = (9/13)^4 = 6561/28561 \approx 23.0\%$$

3. (a): Assume that the probability of a face-down card is E_1 and a face-up is E_2

$$P(E_1|E_2) = P(E_1 \cap E_2) / P(E_2) = 16/51$$

- (b): Assume that the probability of the opponent's face-down card is E_1 and face-up card is ace and your own hand consists of a 9 and a 10 is E_2 .

$$P(E_1|E_2) = P(E_1 \cap E_2) / P(E_2) = 15/49$$

4. (a): $P(A) = 1 - C(n-k, k) / C(n, k)$

- (b): If k is even:

$$P(B) = C(k, k/2) * C(n-k, k-k/2) + C(k, k/2+1) * C(n-k, k-k/2-1) + \dots + C(k, k) * C(n-k, 0) / C(n, k) = \sum_{i=\frac{k}{2}}^k \frac{C_{n-k}^{k-i} C_k^i}{C_n^k}$$

- If k is odd:

$$P(B) = \sum_{i=\frac{k}{2}+0.5}^k \frac{C_{n-k}^{k-i} C_k^i}{C_n^k}$$

5. $P(A) = 3/5 * 3/5 * (C(3, 1) - 1) * 2/5 + 3/5 * 3/5 = 81/125 = 64.8\%$

6. See below:

TIMES	1	2	3	4
P	1/2	1/2 * 1/2	(0.5) ³	(0.5) ³

$$E(\text{TIMES}) = 1 * 1/2 + 2 * 1/4 + 3 * 1/8 + 4 * 1/8 = 15/8$$

7. Assume the probability of choosing chips produced by F2 is A and defective is B.

$$\Pr(A \cap B) = (3/10 * 2/25) / (1/10 * 1/5 + 3/10 * 2/25 + 1/2 * 3/100) = 24/59$$

8. (a): $P(E = \text{both gold}) = 25/40 * 24/39 = 5/13$
 $P(T = \text{one gold and one silver}) = 25/40 * 15/39 = 25/124$
 $P(Y = \text{one silver and one gold}) = 15/40 * 25/39 = 25/104$
 $P(L = \text{both silver}) = 15/40 * 14/39 = 7/52$



(b): $P(B) = 5/39 + 25/104 = 5/13$

(c): $P(O) = 25/104 + 25/104 = 25/52$

9. $P(E) = P(E \cap F) + P(E \cap \overline{F})$

$P(E \cap \overline{F}) = P(E) - P(E \cap F)$

Because E and F are independent: $P(E \cap F) = P(E) * P(F)$

$P(E \cap \overline{F}) = P(E) - P(E) * P(F) = P(E) * (1 - P(F)) = P(E) * P(\overline{F})$

According to the definition of an independent event, we can know that E and $\overline{F}$ are independent events. In the same way, we can find F and $\overline{E}$, $\overline{E}$

and $\overline{F}$ are independent.

10. See below:

VALUE	5	11	12	17	18
P	1/10	2/5	1/5	1/10	1/5

$$E(\text{VALUE}) = 1/2 + 22/5 + 12/5 + 17/10 + 18/5 = 12.6$$

11. $P(A) = 1/3 + 2/3 * 1/3 + 2/3 * 2/3 * 1/3 = 19/27$

12. See below:

MONEY	-3	-2	-1	2	3
P	1/16	$1/16 * C(4, 1) = 1/4$	$1/16 * C(4, 2) = 3/8$	$1/16 * C(4, 3) = 1/4$	1/16

$$E(X) = -3/16 + (-1/2) + (-3/8) + 1/2 + 3/16 = -0.375$$